

[KAN-5] [WebRTC Signaling] P2P Video handshake drops and throws 502 Bad Gateway when camera permissions are toggled mid-flight

Created: 13/Sep/26 Updated: 13/Sep/26 Resolved: 13/Sep/26

Status:	Done
Project:	My Software Team
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Task	Priority:	High
Reporter:	Viinesh Kerisnan	Assignee:	Unassigned
Resolution:	Done	Votes:	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Rank:	0ji00033:
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Description

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During the initialization of a video consultation, if the user revokes and re-grants camera permissions in the browser while the WebRTC signaling sequence is active, the Supabase Edge Function drops the WebSocket connection. This results in a dead state where the UI spins indefinitely, and the doctor never receives the connection offer.

Pre-Conditions:

- Patient is authenticated via JWT.
- Doctor is in the "Waiting Room" state, actively listening on the signaling channel.

Environment:

- Frontend: React.js (v18.2.0)
- Backend: Supabase Edge Functions / PostgreSQL
- Browser: Chrome v117 (Windows 11) & Safari v16 (macOS)
- Network: 5G / High-speed Wi-Fi

Steps to Reproduce:

1. Log in as a Patient and navigate to /consultation/join/88472.
2. Click "Join Video Call" to initiate the WebRTC connection.
3. Immediately (within 500ms), click the browser's URL padlock icon and block the Camera permission.
4. Wait 2 seconds, then re-enable the Camera permission.
5. Observe the UI and the Network tab in Developer Tools.

Expected Result:

The React frontend should catch the permission change, gracefully terminate the initial incomplete WebRTC offer, and automatically attempt a UI prompt to restart the signaling process.

Actual Result:

The UI enters an infinite loading state. The Network tab shows a WebSocket connection drop.

Console Error Trace:

WebSocket connection to 'wss://[PROJECT_ID].supabase.co/functions/v1/webrtc-signal' failed: Error during WebSocket handshake: Unexpected response code: 502

Database Impact:

Checking the Supabase consultation_logs table reveals a null value for connection_established_at, meaning the backend thread is hanging open and consuming resources.

Severity: Critical (Breaks core application functionality; potential memory leak on backend)

Priority: Highest (Requires immediate hotfix)

Attachments: network-trace-webrtc-fail.har (Simulated), screen-recording.mp4 (Simulated)

Comments

Comment by Viinesh Kerisnan [13/Sep/26]
Good catch. The issue is a race condition in the React component lifecycle. When the permissions change, the component re-renders and fires a second signaling offer before the Edge Function can clean up the first one. I'm going to add a debounce function to the permission listener and implement a hard timeout on the backend WebSocket.
Comment by Viinesh Kerisnan [13/Sep/26]
Verified on staging environment. The debounce function prevents the double-fire. Toggling permissions mid-flight now correctly throws a local UI error modal asking the patient to refresh, and the backend WebSocket safely terminates with a 1000 Normal Closure code. Memory leak mitigated. Closing ticket.

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